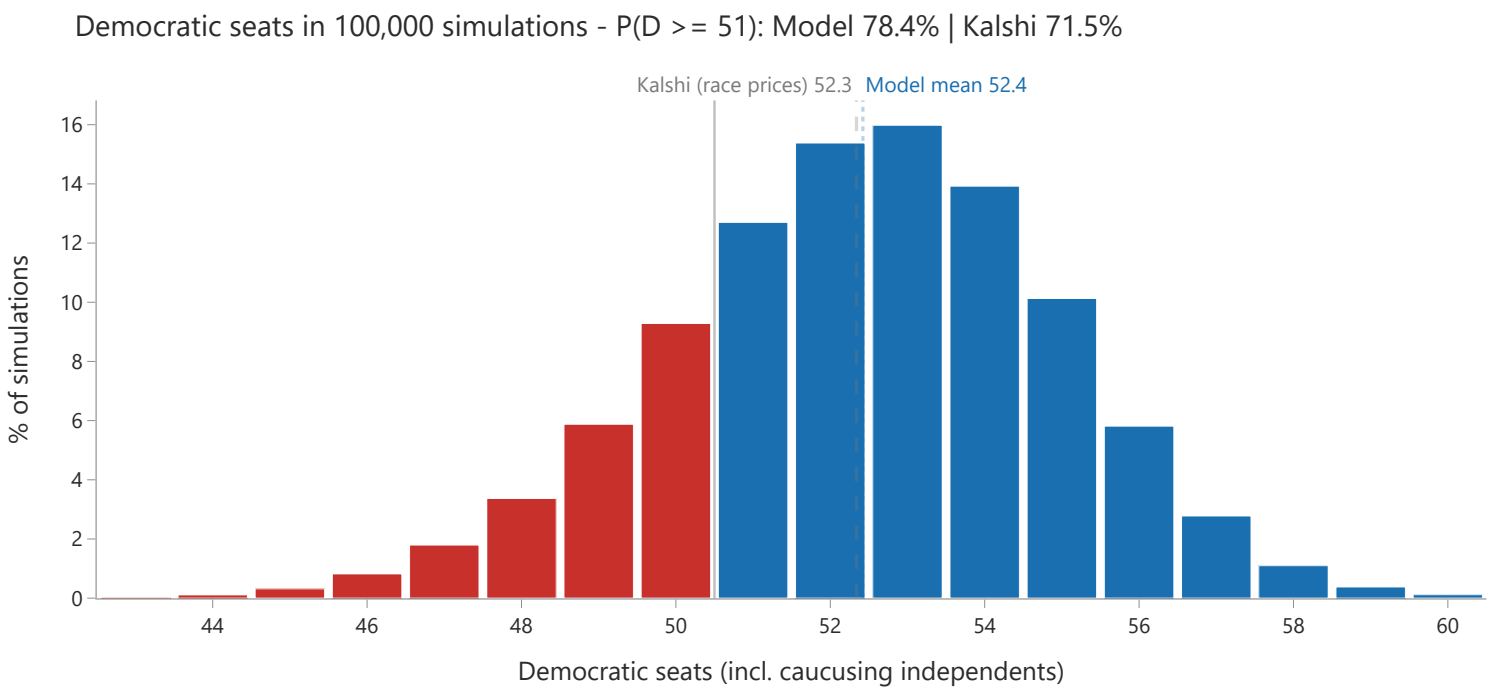
# 2026 Senate forecast - Most_Magnificent_Model scheduled

Team 5: Elizabeth Miller, Mark Scott, Ethan Verrone, Boaz Alkalay, Leo Westebbe
scheduled run, 2026-10-31 11:10

| P(DEMOCRATS >= 51 SEATS) | EXPECTED DEMOCRATIC SEATS | KALSHI CONTROL MARKET | RUN |
|---|---|---|---|
| **78.4%** | **52.4** | **71.5%** | **2026-10-31 11:10** |
| simulation precision ±0.3 pts | median 53 \| 80%: 49-56 \| 95%: 47-57 | sum of race prices: 52.3 seats | 100,000 simulations \| data as of 2026-10-31 \| 5 min |

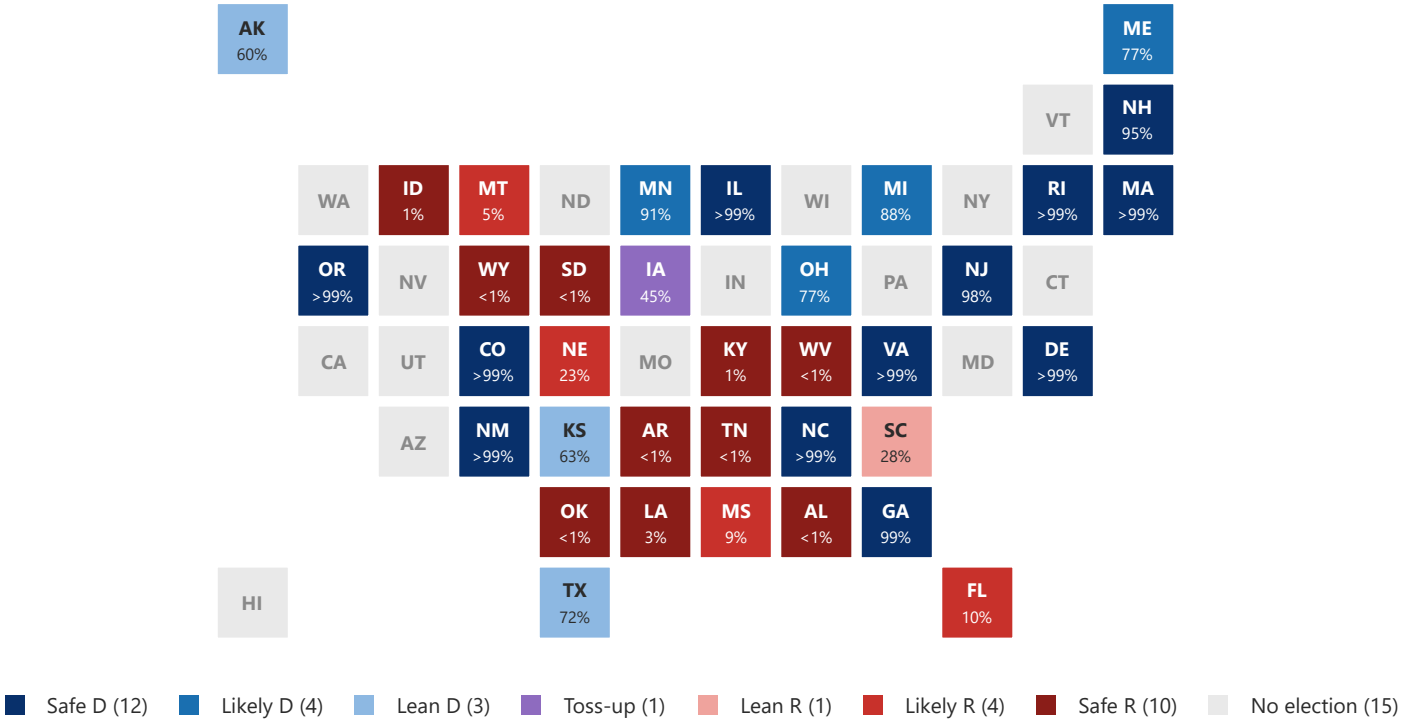
The model and Kalshi expect about the same number of Democratic seats (52.4 vs 52.3); the model's higher control odds come from rating Kansas, Michigan, Maine and Ohio higher for Democrats than the market does.

## Seat distribution

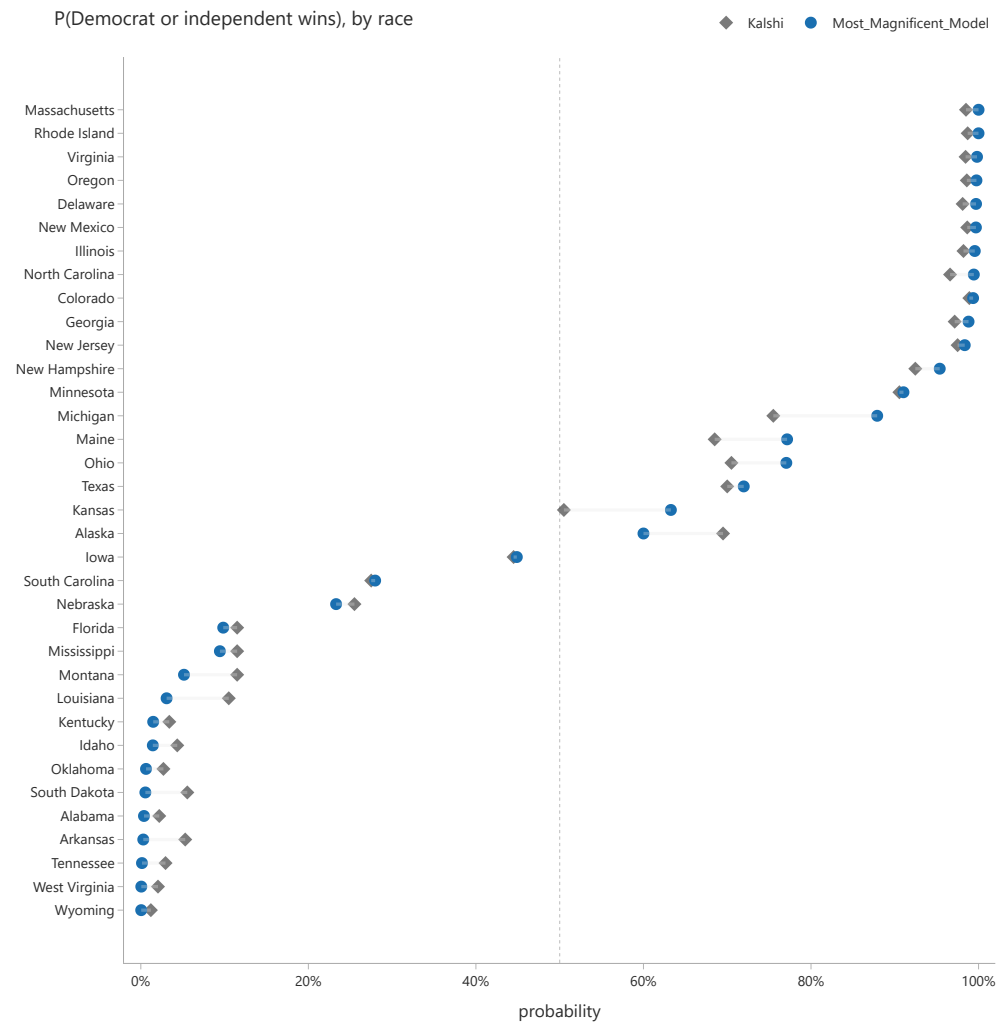


## Map

Race ratings (number = Model P(D/Ind wins); D includes independents)
Toss-up under 60%, Lean 60%-75%, Likely 75%-95%, Safe 95%+



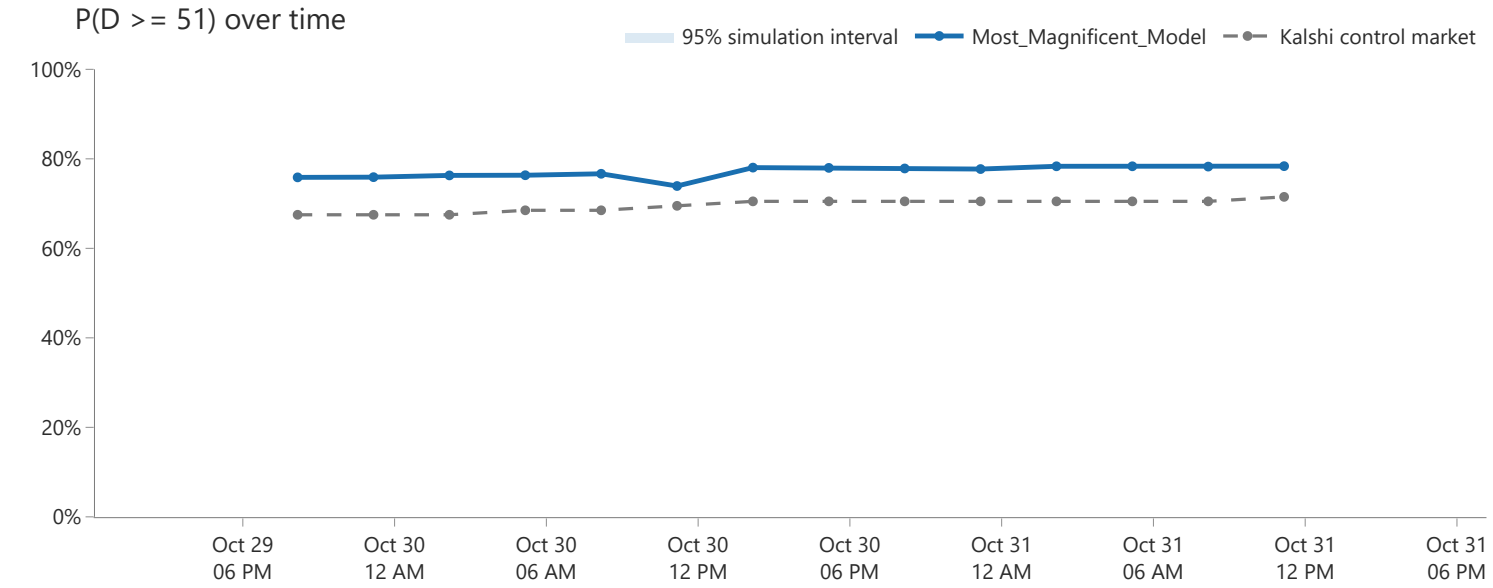
## Race by race

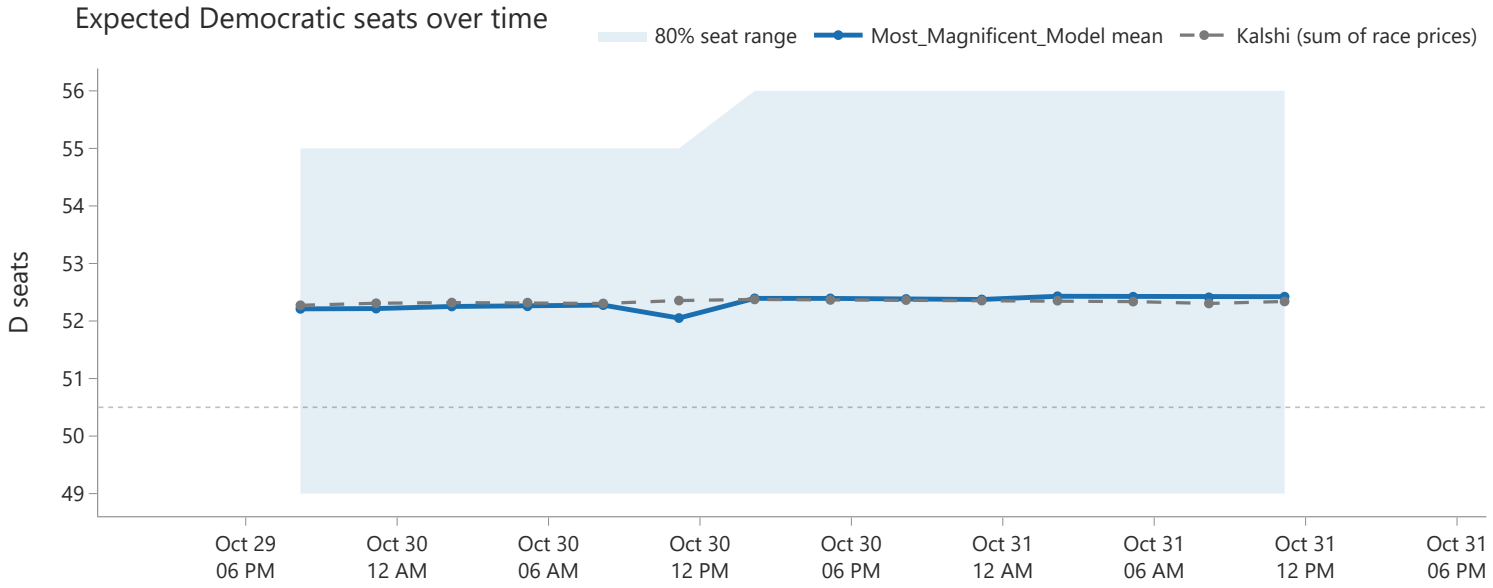



| Race | Call | Model P(D/Ind) | Kalshi P(D/Ind) | Model - Kalshi (pts) | Projected margin | +/- (1 SD) | Polls | Model change since first run (pts) | Kalshi change since first run (pts) |
|---|---|---|---|---|---|---|---|---|---|
| Massachusetts | Safe D | >99% | 98% | +2 | D+34.2 | +/-5.7 | 13 | +0 | +0 |
| Rhode Island | Safe D | >99% | 99% | +1 | D+25.4 | +/-5.5 | 4 | +0 | -0 |
| Virginia | Safe D | >99% | 98% | +1 | D+18.8 | +/-5.9 | 1 | +0 | -0 |
| Oregon | Safe D | >99% | 99% | +1 | D+28.7 | +/-9.4 | 0 | +0 | -0 |
| Delaware | Safe D | >99% | 98% | +2 | D+28.7 | +/-9.4 | 0 | -0 | -0 |
| New Mexico | Safe D | >99% | 99% | +1 | D+18.8 | +/-6.4 | 2 | +0 | +0 |
| Illinois | Safe D | >99% | 98% | +1 | D+14.7 | +/-5.3 | 3 | -0 | +0 |

| Race | Call | Model P(D/Ind) | Kalshi P(D/Ind) | Model - Kalshi (pts) | Projected margin | +/- (1 SD) | Polls | Model change since first run (pts) | Kalshi change since first run (pts) |
|---|---|---|---|---|---|---|---|---|---|
| North Carolina | Safe D | >99% | 97% | +3 | D+12.3 | +/-4.7 | 40 | +0 | +0 |
| Colorado | Safe D | >99% | 99% | +0 | D+24.7 | +/-9.4 | 0 | +0 | +1 |
| Georgia | Safe D | 99% | 97% | +2 | D+11.0 | +/-4.7 | 18 | +0 | +1 |
| New Jersey | Safe D | 98% | 98% | +1 | D+20.7 | +/-9.4 | 0 | -0 | +0 |
| New Hampshire | Safe D | 95% | 92% | +3 | D+8.4 | +/-5.0 | 21 | +0 | +2 |
| Minnesota | Likely D | 91% | 91% | +0 | D+6.3 | +/-4.8 | 7 | +1 | +1 |
| Michigan | Likely D | 88% | 76% | +12 | D+5.3 | +/-4.6 | 43 | +0 | +1 |
| Maine | Likely D | 77% | 68% | +9 | D+3.2 | +/-4.5 | 18 | +2 | +5 |
| Ohio | Likely D | 77% | 70% | +7 | D+3.3 | +/-4.6 | 32 | +1 | +1 |
| Texas | Lean D | 72% | 70% | +2 | D+2.4 | +/-4.4 | 53 | +3 | +1 |
| Kansas | Lean D | 63% | 50% | +13 | D+1.5 | +/-4.6 | 7 | +16 | +2 |
| Alaska | Lean D | 60% | 70% | -9 | D+1.1 | +/-4.6 | 18 | +1 | +1 |
| Iowa | Toss-up | 45% | 44% | +0 | R+0.6 | +/-4.7 | 24 | -0 | -3 |
| South Carolina | Lean R | 28% | 28% | +0 | R+2.7 | +/-4.7 | 4 | -0 | +0 |
| Nebraska | Likely R | 23% | 26% | -2 | R+3.6 | +/-5.0 | 4 | -2 | -7 |
| Florida | Likely R | 10% | 11% | -2 | R+5.8 | +/-4.6 | 17 | -0 | +1 |
| Mississippi | Likely R | 9% | 11% | -2 | R+7.1 | +/-5.4 | 3 | +0 | +0 |
| Montana | Likely R | 5% | 11% | -6 | R+9.3 | +/-5.8 | 26 | +1 | -1 |
| Louisiana | Safe R | 3% | 10% | -7 | R+13.2 | +/-7.0 | 0 | +0 | +1 |
| Kentucky | Safe R | 1% | 3% | -2 | R+21.1 | +/-9.4 | 0 | -0 | -0 |
| Idaho | Safe R | 1% | 4% | -3 | R+15.3 | +/-6.8 | 1 | -0 | -1 |
| Oklahoma | Safe R | <1% | 3% | -2 | R+25.1 | +/-9.4 | 0 | -0 | +0 |
| South Dakota | Safe R | <1% | 6% | -5 | R+13.8 | +/-5.1 | 4 | +0 | +0 |
| Alabama | Safe R | <1% | 2% | -2 | R+16.2 | +/-5.7 | 5 | -0 | +0 |
| Arkansas | Safe R | <1% | 5% | -5 | R+14.5 | +/-5.0 | 5 | -0 | +1 |
| Tennessee | Safe R | <1% | 3% | -3 | R+22.9 | +/-6.8 | 1 | -0 | -1 |
| West Virginia | Safe R | <1% | 2% | -2 | R+36.9 | +/-9.4 | 0 | -0 | +0 |
| Wyoming | Safe R | <1% | 1% | -1 | R+37.1 | +/-9.4 | 0 | -0 | +0 |

## History


P(D >= 51) over time
95% simulation interval
Most_Magnificent_Model
Kalshi control market
100%
80%
60%
40%
20%
0%
Oct 29
06 PM
Oct 30
12 AM
Oct 30
06 AM
Oct 30
12 PM
Oct 30
06 PM
Oct 31
12 AM
Oct 31
06 AM
Oct 31
12 PM
Oct 31
06 PM


Expected Democratic seats over time
80% seat range
Most_Magnificent_Model mean
Kalshi (sum of race prices)
D seats
56
55
54
53
52
51
50
49
Oct 29
06 PM
Oct 30
12 AM
Oct 30
06 AM
Oct 30
12 PM
Oct 30
06 PM
Oct 31
12 AM
Oct 31
06 AM
Oct 31
12 PM
Oct 31
06 PM

## Notes

- Model Testing: 35 PASS | 6 WARN | 0 FAIL (details in model_warnings.xlsx).
- Data: votehub_senate live, votehub_generic_ballot live, votehub_approval live, kalshi_races live, kalshi_national live.

## Key settings

| | |
|---|---|
| Poll source | VoteHub API |
| Poll age | all polls, half-life 7 days |
| Polls : fundamentals : Kalshi | Bayesian precision (poll floor 3.9, prior SD 9.3); Kalshi precision 1/8^2, longshot-corrected, conversion SD >= 4.42 |
| Poll reliability (thin races) | automatic (Bayesian precision) |
| Fundamentals | PVI x 2 + national environment (generic-ballot blend 0.75) + candidate effects x 0.30 |
| Incumbency | 3.8 pts (appointed 1) |
| LV : RV : A | 1 : 0.85 : 0.7; RV shift +0.9 (measured) |
| Flooding penalty | 1/k^0.5 |
| House effects | in-cycle, shrunk to scorecard (k 5) |
| Ungraded pollsters | random |
| National signals | VoteHub polls + Kalshi averages (= 5 polls) |
| Kalshi races | Bayesian input + comparison (found automatically) |
| Source errors | correlated (rho 0.3); sims: national 2.5 + region 1.5 + state, in pts |
| Calibration | calibration.json (2026-10-29 22:45) |
| Simulations | 100,000 |

## Equations and weights

1. WEIGHT OF EACH POLL

$$w = 0.5^{\text{age}/7} \times \left(\frac{\min(n,\,2500)}{800}\right)^{0.5} \times w_{\text{pop}} \times w_{\text{partisan}} \times w_{\text{generic}} \times w_{\text{field}} \times w_{\text{quality}} \times w_{\text{accuracy}} \times \frac{1}{k^{0.5}} \times \frac{1}{\text{versions}}$$

Population: LV 1, RV 0.85, adults 0.7. Partisan sponsor 0.6. Generic-party question 0.5. Full field 0.9. Quality 0.5 to 1 by VoteHub grade (ungraded firms get a random graded firm's grade in each simulation). Accuracy = (1.25 x 4 / poll SD)^2, kept between 0.5 and 2. k = the pollster's polls of that race.

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2. EACH POLL'S MARGIN

$$m = (D - R)\,\frac{D + R + U}{D + R} + s_{RV} \pm s_{\text{partisan}} - h, \qquad h = \frac{\sum \text{residuals} + 5\,h_{\text{scorecard}}}{n_{\text{polls}} + 5}$$

U = undecided. RV shift s_RV = +0.88 pts (registered-voter polls only; measured from LV/RV pairs). Partisan shift = 3 pts against the sponsor. Residual = poll minus the same state's average of other pollsters, from the last 60 days.

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3. POLLING AVERAGE

$$\bar{m} = \frac{\sum w\,m}{\sum w}, \qquad SD_{\text{polls}} = \sqrt{\frac{SD_{\text{poll}}^2}{n_{\text{eff}}} + 3.87^2 + SD_U^2}, \qquad n_{\text{eff}} = \frac{(\sum w)^2}{\sum w^2}$$

The middle term is the error all polls of a state share, so more polls can't remove it.

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4. NATIONAL ENVIRONMENT

$$\text{env} = 0.75 \times \text{GB} + 0.25 \times 0.35 \times (-\text{net approval}), \qquad \text{signal} = \frac{n_{\text{fresh}} \cdot \text{VoteHub} + 5 \cdot \text{Kalshi}}{n_{\text{fresh}} + 5}$$

GB = generic ballot. Each VoteHub poll counts 0.5^(age / 14) toward n_fresh. This run: env = D+8.9.

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5. PRIOR (FUNDAMENTALS)

$$\text{prior} = 2 \times \text{PVI} + \text{env} \pm 3.78\,[\text{incumbent}] + 0.30 \times c, \qquad SD_{\text{prior}} = 9.32$$

Appointed incumbents get 1 pt. c = candidate effect: the nominees' past margin minus a generic candidate's (capped at +/- 25). Independent-led races: SD x 1.3.

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6. KALSHI

$$P' = \text{logistic}\,(1.15 \cdot \text{logit}\,P), \qquad \text{margin}_K = \Phi^{-1}(P') \cdot \max\,(4.42,\ SD_{F+P}), \qquad SD_K = 8$$

Used only when 6% <= P <= 94%.

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7. EACH RACE

$$\text{margin} = \frac{\pi_F \cdot \text{prior} + \pi_P \cdot \bar{m} + \pi_K \cdot \text{margin}_K}{\pi_F + \pi_P + \pi_K}, \quad \pi = \frac{1}{SD^2}, \quad SD^2 = \frac{1}{\Pi} + \frac{2(0.3)}{\Pi^2}\left(\sqrt{\pi_F\pi_P} + \sqrt{\pi_F\pi_K} + \sqrt{\pi_P\pi_K}\right) + 0.76^2 \cdot \text{days left}$$

Pi = the sum of the three precisions. The second term widens the range because the three sources tend to miss together.

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$$\tilde{m}_i = \text{margin}_i + N + R_{g(i)} + L\,f_i + e_i, \qquad \text{D seats} = 34 + \sum_i \mathbf{1}[\tilde{m}_i > 0], \qquad P(D \geq 51) = \text{share of simulations}$$

N = national miss (SD 2.52, t with df 8). R = regional miss (SD 1.51). L = lean factor (SD 1), f_i = PVI / 10. e = the rest of the race's SD (t with df 10.9). An independent's win counts for Democrats with probability: Nebraska 0.8.